Name:_____ Period:_____
Earth and Space Science

SEDIMENTARY ROCKS
LEARNING GUIDE

Learning Intention
Today we will be learning how sediments and dissolved minerals become sedimentary rocks so we can explain how the striped layers of Badlands National Park and the gypsum of White Sands formed.

Success Criteria
I can… ● Describe the steps that turn sediment into sedimentary rock, from weathering to cementation. ● Classify sedimentary rocks as clastic, chemical, or organic and explain how each one forms. ● Explain how sediments settle into sorted layers and why fossils are found in sedimentary rock.

Badlands and White Sands: Rock Built Layer by Layer

Badlands National Park, in South Dakota, is famous for its striped buttes and spires. Each stripe is a layer of sediment, such as mud, sand, or volcanic ash, that settled on top of the layer before it over millions of years. On Day 1, you saw these layers at Site C. Far to the south, at White Sands National Park in New Mexico, the dunes are made of gypsum crystals that formed when water evaporated. In this guide, these parks are our examples of the two main ways sedimentary rock forms. As you read, keep this question in mind: *How do loose grains and dissolved minerals become solid rock?*

From Sediment to Rock

Sedimentary rocks form at or near Earth's surface. Most form in five steps:

- **Weathering:** Rock at the surface is broken into smaller pieces called **sediment**.
- **Erosion:** Water, wind, or ice carries the sediment away.
- **Deposition:** When the water or wind slows down, it loses energy and drops the sediment.
- **Compaction:** As layers pile up, their weight squeezes the sediment together and presses out water.
- **Cementation:** Minerals dissolved in the water, such as calcite or quartz, crystallize between the grains and glue them together into solid rock.

Check for Understanding

1. Number the steps **1** through **5** in the order they happen.

_____ compaction

_____ weathering

_____ cementation

_____ deposition

_____ erosion

_____ 2. Which process glues sediment grains together into solid rock?
(1) weathering
(2) erosion
(3) compaction
(4) cementation

Three Kinds of Sedimentary Rock

- **Clastic** rocks are made of fragments of other rocks and are named by grain size.
- **Chemical** rocks form when minerals crystallize out of water, often as it evaporates. Rocks that form when water evaporates are called **evaporites**.
- **Organic** rocks form from the remains of living things.

Kind	Rock	What it is made of
Clastic	Conglomerate	Rounded pebbles cemented together
Clastic	Sandstone	Sand grains
Clastic	Siltstone	Silt (finer than sand)
Clastic	Shale	Clay (the finest sediment); splits into thin layers
Chemical	Rock salt	Halite crystals left when salt water evaporates
Chemical	Rock gypsum	Gypsum crystals left when water evaporates (White Sands)
Organic	Limestone	Shells and skeletons of sea animals, made of calcite
Organic	Coal	Compacted remains of plants

Check for Understanding

1. Write **CL** (clastic), **CH** (chemical), or **O** (organic) for each rock.

   _____ a. Sandstone

   _____ b. Rock gypsum

   _____ c. Coal

   _____ d. Conglomerate

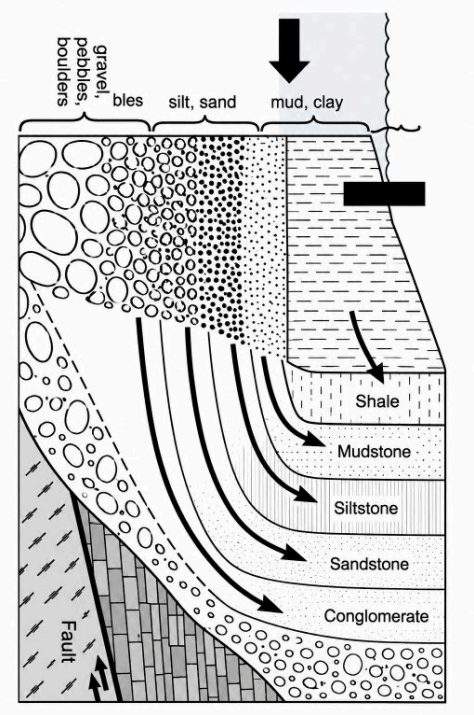
2. Explain why the gypsum at White Sands is classified as a chemical sedimentary rock.

_____

_____

ESSRT Connection

Use the **sedimentary rocks section** of the **Rock Cycle Infographic** (page 15) in the 2026 Revised Edition Reference Tables for Earth and Space Sciences.



_____ 1. According to the infographic, which rock forms from the largest sediments?
(1) shale
(2) siltstone
(3) sandstone
(4) conglomerate

2. According to the infographic, which rock forms from clay, the smallest sediment?

_____

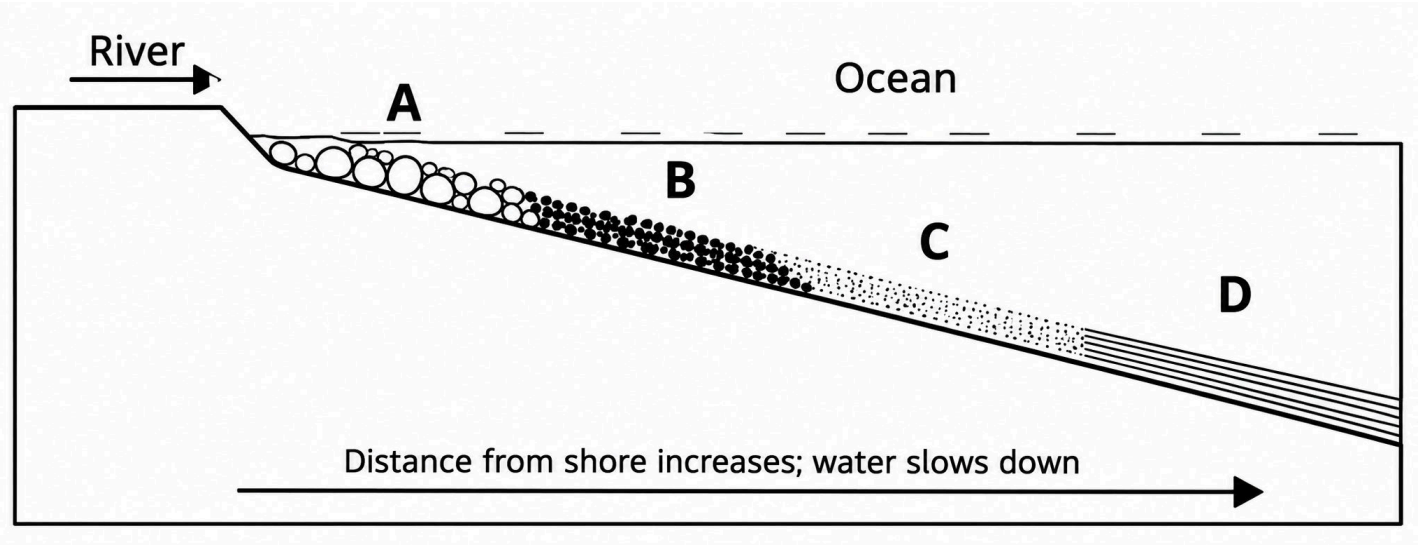
Layers, Sorting, and Fossils

When a river flows into a lake or ocean, the water slows down and loses energy. The **largest, heaviest** sediments settle out first, close to shore. Smaller and smaller sediments are carried farther out before they settle, and the tiny clay particles settle last, in the calmest water. This pattern is called **horizontal sorting**. It explains why conglomerate forms near shore and shale forms in deep, quiet water.

Sediment settles in flat layers called **strata**, with each new layer on top of older ones. The stripes of the Badlands are layers like this. They are different from the **bands** in gneiss, which are minerals that separated during metamorphism. Sedimentary rocks are also where most **fossils** are found, because sediment can bury the remains of living things gently without destroying them. You will use fossils to study Earth's history in Unit 9.

Check for Understanding: Model It

The diagram represents sediments settling where a river enters the ocean.



_____ 1. At which location are the largest sediments deposited?
(1) A
(2) B
(3) C
(4) D

_____ 2. The sediment at location D would most likely become which rock?
(1) conglomerate
(2) sandstone
(3) siltstone
(4) shale

3. A student says the stripes in the Badlands are bands, like in gneiss. Explain why the student is incorrect.

_____

_____

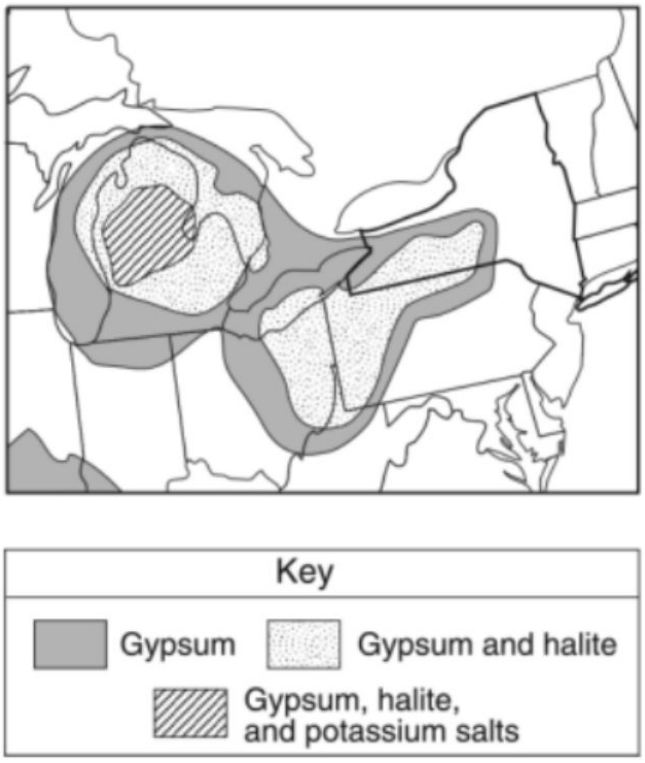
Cluster Set 1

Base your answers to questions 1 through 3 on the passage and map below and on your knowledge of Earth and Space Science.

Salt Beneath New York

Hundreds of millions of years ago, shallow seas covered much of what is now New York and the Great Lakes region. In a warm, dry climate, the seawater evaporated again and again, leaving behind layers of gypsum and halite. Today, salt is mined from these layers deep beneath central New York. The map shows where these mineral deposits are found.

Mineral Deposits



_____ 1. The halite deposits shown on the map form which sedimentary rock?
(1) limestone
(2) rock salt
(3) shale
(4) conglomerate

_____ 2. These deposits formed mainly by
(1) cooling of magma
(2) compaction of pebbles
(3) evaporation of seawater
(4) compaction of plant remains

3. The gypsum at White Sands and the salt under New York are both evaporites. Explain what both locations must have had in common when these rocks formed.

_____

_____

Mini Glossary

Cementation – the gluing of sediment grains together by minerals that crystallize from water

Chemical sedimentary rock – a rock formed when minerals crystallize out of water (examples: rock salt, rock gypsum)

Clastic sedimentary rock – a rock made of fragments of other rocks, named by grain size

Compaction – the squeezing together of sediment by the weight of layers above it

Deposition – the dropping of sediment when water, wind, or ice slows down

Erosion – the carrying away of sediment by water, wind, or ice

Evaporite – a chemical sedimentary rock formed when water evaporates

Fossil – the preserved remains or traces of a living thing

Horizontal sorting – the pattern of larger sediments settling first and smaller ones settling farther away

Organic sedimentary rock – a rock formed from the remains of living things (examples: limestone, coal)

Sediment – loose pieces of rock, mineral, or shell

Sedimentary rock – a rock formed from sediments or from minerals that crystallize out of water

Strata – layers of sedimentary rock

Weathering – the breaking down of rock into smaller pieces